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SCUOLA DI INGEGNERIA INDUSTRIALE
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Assignment 1
Group 12

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1. Introduction

In this project, we address the problem of client segmentation in the financial industry using a data-driven approach. Traditional segmentation methods often rely on simplified categorizations that fail to capture the heterogeneity of customer profiles. To overcome this limitation, we employ unsupervised machine learning techniques on a dataset combining demographic, financial, behavioral, and categorical variables.
The objective is to identify meaningful groups of clients, called personas, that can support more effective financial decision-making and personalized service strategies.

2. Data Preprocessing and Feature Engineering

The dataset was first loaded and inspected, removing non-informative variables such as client identifiers. Given the heterogeneous nature of the data, features were divided into three main categories: categorical variables, demographic numerical variables, and financial/behavioral numerical variables.
Categorical features were transformed through one-hot encoding to allow their inclusion in distance computations. Demographic variables were scaled to ensure comparability across features, while financial variables, already normalized in the $[0, 1]$ range, were left unchanged to preserve their economic interpretation.
All processed features were then combined into a single feature matrix. Additional exploratory analysis (histograms, boxplots, and categorical distributions) confirmed that the dataset is well-behaved, with no significant outliers or distortions, ensuring its suitability for clustering. One notable insight is the strong concentration of clients in the `Employee/Worker` category and in the `North` area, which could be relevant when interpreting the final clusters.

3. Distance Design for Heterogeneous Data

A key challenge of this project is defining an appropriate notion of similarity for mixed-type data. Standard distances such as Euclidean or Manhattan are not suitable, as they fail to properly account for categorical variables and heterogeneous scales.
To address this, we adopted a feature-level weighted distance inspired by the Gower metric. Each variable contributes directly to the overall distance with a specific weight, while preserving normalization across variable types. Formally, the distance between two observations is defined as:

$$D(i, j) = \sum_k w_k \, d_k(i, j) \tag{1}$$

where ordinal variables are represented through normalized rank differences ($\frac{|r_{ik}-r_{jk}|}{R_k}$), numerical variables are measured using scaled absolute differences ($\frac{|x_{ik}-x_{jk}|}{R_k}$), and financial variables are modeled via weighted Manhattan contributions ($w_k|x_{ik} - x_{jk}|$).
This approach avoids information loss due to aggregation and allows us to emphasize financially relevant features, which are the primary drivers of client differentiation.

4. Limitations of Linear Dimensionality Reduction

To explore the structure of the data, we first applied **Principal Component Analysis**. However, the first three components explain less than half of the total variance, and no clear cluster structure emerges in low-dimensional projections.
We further applied **Factor Analysis** and **Independent Component Analysis**. While FA identifies a dominant latent factor related to financial and behavioral variables, the overall explanatory power remains limited. ICA, on the other hand, produces fragmented and difficult-to-interpret representations.
These results suggest that the dataset cannot be effectively summarized through linear or low-dimensional latent structures, motivating the use of nonlinear and distance-based methods.

5. Nonlinear Embeddings and Distance Validation

To evaluate the effectiveness of the proposed distance, we used nonlinear embedding techniques such as **t-SNE** and **UMAP**.
When standard distances are applied, both methods produce scattered and weakly structured embeddings, indicating poor representation of similarities. In contrast, using the custom weighted distance leads to more coherent and compact structures, with emerging cluster patterns. For t-SNE, the custom weighted distance yielded more coherent patterns and clearer cluster organization, especially for intermediate perplexity values, confirming its superior ability to capture meaningful similarities. As for UMAP, when given the custom weighted distance, it provides an even more stable representation, highlighting the presence of a limited number of clusters, roughly between two and six.

6. Clustering and Model Selection

Clustering was performed using the **K-medoids** algorithm on the custom distance matrix. This method is particularly suitable, as it does not rely on Euclidean geometry and is robust to outliers.
The results reveal a clear but coarse structure: a strong separation is observed for $k = 2$, while $k = 3$ introduces a meaningful additional subdivision. Increasing the number of clusters beyond this point mainly leads to fragmentation rather than new informative groups.
To select the optimal number of clusters, we used three evaluation metrics: *Calinski-Harabasz*, *Davies-Bouldin*, and *Silhouette score*. The combined evidence suggests that $k = 3$ provides the best trade-off between separation and stability.
Hierarchical clustering further confirms this result, showing a three-branch structure consistent with previous findings.
In particular, one branch appears well separated at higher linkage distances, while the remaining observations form two substructures that merge at lower distances. This hierarchical view is consistent with previous findings: a strong primary split, captured by $k = 2$, and a secondary subdivision, supporting $k = 3$.

7. Cluster Interpretation

The final clusters were interpreted by analyzing summary statistics and variable distributions.

- **Cluster 0**: represents a high-value segment, characterized by higher income, wealth, financial literacy, and digital engagement. This group corresponds to financially sophisticated and proactive clients.
- **Cluster 1**: includes individuals with lower income, wealth, and financial engagement, representing a more fragile or less active segment. It represents the elderly clients.
- **Cluster 2**: lies in between, with moderate characteristics and more balanced profiles, capturing a broad and heterogeneous middle segment, where the main characteristic is a bigger mean family composition.

Overall, the segmentation reflects gradual differences in financial behavior rather than sharply separated groups, as expected from our starting dataset.

8. Bayesian Extension and Dynamic Profiling

To extend the static clustering framework to a dynamic one, we implemented a Bayesian classification approach for assigning new clients to clusters. Using cluster-specific distributions and prior probabilities, posterior probabilities were computed to quantify assignment uncertainty.
Synthetic clients were generated via bootstrap, with small Gaussian perturbations, and assigned to clusters, showing that the intermediate cluster is the most populated, while extreme profiles remain more selective.
We then updated cluster profiles using a Bayesian updating procedure, which refines parameter estimates and reduces uncertainty. This results in more stable and informative personas, while preserving their economic interpretation:

- **Cluster 0**: reinforced as a high-value, digitally engaged segment, with increases in income, wealth, and financial sophistication.
- **Cluster 1**: confirmed as a low-engagement and less sophisticated segment, with generally lower financial indicators but slightly stronger ESG orientation.
- **Cluster 2**: consolidated as an intermediate and balanced segment, with moderate financial characteristics and relatively higher saving behavior.

9. Learning Vector Quantization

Finally, we applied **Learning Vector Quantization** to learn representative prototypes for each cluster. This is extremely useful since it allows us to move from simple clustering to a reusable classification framework for new clients. The model was first trained on synthetic data generated from cluster distributions and then used to classify real observations.
The results show strong agreement with the original clustering, confirming that the learned prototypes effectively capture the underlying structure.

10. More Emphasis on Age

The initial clustering showed a key limitation: age had little impact on the segmentation, leading to clusters with similar and unrealistically high average ages. This was due to its low weight in the distance function.
We address this by introducing a dedicated `AgeBin` block in the distance function, giving age an explicit and undiluted weight. We also revisited the number of clusters, setting $k = 4$ and assigned the age a higher weight. The quantitative metrics are slightly worse, as reflected by increased overlap and a lower Silhouette score, but the segmentation is more meaningful from a business perspective. Overall, this highlights a trade-off between numerical performance and practical relevance.

11. Conclusion

Given all the studies, we can say that the proposed weighted distance successfully captures the structure of the data, enabling meaningful clustering and interpretation.
The combination of clustering, nonlinear embeddings, and Bayesian updating provides a flexible and robust framework for both static segmentation and dynamic client classification, with clear applications in financial decision-making.